CM0832 - MT5001 Elements of Set Theory
2. Relations, Functions and Orderings

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Outline

Outline

Preliminaries

Textbook

Karel Hrbacek and Thomas Jech ([1978] 1999). Introduction to Set Theory.

Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

Introduction

Starting...

> *"We begin our program of developing set theory as a foundation for mathematics by showing how various general mathematical concepts, such as relations, functions, and orderings can be represented by sets." (p. 17)*

Outline

Ordered Pairs

Definition [2.]1.1

Let a and b be sets. We define the **ordered pair** of a and b, denoted (a, b), where a is the **first coordinate** and b is the **second coordinate**, using Kuratowski (1921)'s definition, that is,

$$(a, b) \stackrel{\text{def}}{=} \{\{a\}, \{a, b\}\}.$$

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Question

Is (a, b) a set?

Ordered Pairs

Example

We show that $(\emptyset, \{\emptyset\}) \doteq\!\!\!\!\!/\ (\{\emptyset\}, \emptyset)$.

$$(\emptyset, \{\emptyset\}) \overset{\text{def}}{=} \{\, \{\emptyset\}, \{\emptyset, \{\emptyset\}\} \,\} \not\doteq \{\, \{\{\emptyset\}\}, \{\{\emptyset\}, \emptyset\} \,\} \overset{\text{def}}{=} (\{\emptyset\}, \emptyset).$$

Ordered Pairs

Example

Let a be a set. Then

$$
\begin{aligned}
(a, a) &\stackrel{\text{def}}{=} \{\{a\}, \{a, a\}\} \\
&\doteq \{\{a\}, \{a\}\} \\
&\doteq \{\{a\}\}.
\end{aligned}
$$

Ordered Pairs

Exercise [2.]1.1

Let a and b sets.

(a) Prove that $(a, b) \in \mathcal{P}(\mathcal{P}(\{a, b\}))$.

(b) Prove that $a, b \in \bigcup(a, b)$.

(c) Prove that if $a \in A$ and $b \in A$, then $(a, b) \in \mathcal{P}(\mathcal{P}(A))$.

Ordered Pairs

Theorem [2.]1.2

Let a, a', b, b' be sets. Then,

$$(a,b) \doteq (a',b') \quad \text{if and only if} \quad a \doteq a' \text{ and } b \doteq b'.$$

Ordered Pairs

Theorem [2.]1.2

Let a, a', b, b' be sets. Then,

$$(a, b) \doteq (a', b') \quad \text{if and only if} \quad a \doteq a' \text{ and } b \doteq b'.$$

Exercise

To give a different definition of ordered pair. Note that your definition must be a set and it must satisfies Theorem [2.]1.2.

Ordered Pairs

Definitions (p. 18)

Let a, b, c and d sets. Then

$$(a, b, c) \stackrel{\text{def}}{=} ((a, b), c) \qquad \textbf{(ordered triples)}$$

$$(a, b, c, d) \stackrel{\text{def}}{=} ((a, b, c), d) \qquad \textbf{(ordered quadruples)}$$

$$(a) \stackrel{\text{def}}{=} a \qquad \textbf{(one-tuples)}$$

Outline

Relations

Definition [2.]2.1

A set R is a **binary relation** if all elements of R are ordered pairs.

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Remark

> *"A binary relation is, therefore, determined by specifying all ordered pairs of objects in that relation; it does not matter by what property the set of these ordered pairs is described." (p. 19)*

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> *"A binary relation is, therefore, determined by specifying all ordered pairs of objects in that relation; it does not matter by what property the set of these ordered pairs is described." (p. 19)*

Notation

Let R be a binary relation. We write $(a, b) \in R$ or aRb.

Relations

Example (informal)

Whiteboard.

Relations

Example (informal)

Whiteboard.

Example (informal)

Let $\mathbf{N} = \{0, 1, 2, \dots\}$. The identity relation on $\mathbf{N}$, denoted $\mathrm{Id}_{\mathbf{N}}$, is defined by

$$\begin{aligned}\mathrm{Id}_{\mathbf{N}} &= \{\, (n, n) \mid n \in \mathbf{N} \,\} \\ &= \{(0,0), (1,1), (2,2), \dots\}.\end{aligned}$$

Definition [2.]2.3

Let R be a binary relation. Then

$$\operatorname{dom} R \stackrel{\text{def}}{=} \{\, x \mid \text{there exits } y \text{ such as } xRy \,\} \qquad (\textbf{domain} \text{ of } R)$$

$$\operatorname{ran} R \stackrel{\text{def}}{=} \{\, y \mid \text{there exits } x \text{ such as } xRy \,\} \qquad (\textbf{range} \text{ of } R)$$

$$\operatorname{field} R \stackrel{\text{def}}{=} \operatorname{dom} R \cup \operatorname{ran} R \qquad (\textbf{field} \text{ of } R)$$

Relations

Definition [2.]2.3

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$$\operatorname{field} R \stackrel{\text{def}}{=} \operatorname{dom} R \cup \operatorname{ran} R \qquad \text{(\textbf{field} of } R\text{)}$$

If $\operatorname{field} R \subseteq X$ then R **is a relation in** X.

Relations

Question

Let R be a binary relation. Are $\operatorname{dom} R$ and $\operatorname{ran} R$ sets?

Relations

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Let R be a binary relation. Are $\operatorname{dom} R$ and $\operatorname{ran} R$ sets?

Answer

Definition of $\operatorname{dom} R$ and $\operatorname{ran} R$ without using the textbook conventions.

$$\operatorname{dom} R \stackrel{\text{def}}{=} \left\{ x \in \bigcup\bigcup R \;\middle|\; \text{there exits } y \text{ such as } xRy \right\},$$
$$\operatorname{ran} R \stackrel{\text{def}}{=} \left\{ y \in \bigcup\bigcup R \;\middle|\; \text{there exits } x \text{ such as } xRy \right\}.$$

That is, $\operatorname{dom} R$ and $\operatorname{ran} R$ are sets defined from the Axiom Scheme of Comprehension.

Relations

Convention (p. 21)

When defining sets of ordered pairs we shall use the following convention:

$$\{\,(x,y) \mid \mathbf{P}(x,y)\,\} \stackrel{\text{def}}{=} \{\,w \mid w \doteq (x,y) \text{ for some } x \text{ and } y \text{ such that } \mathbf{P}(x,y)\,\}.$$

Definition [2.]2.11

Let A be a set.

- The **membership relation** on A, denoted $\in_A$, is defined by

$$\in_A \stackrel{\text{def}}{=} \{\, (a, b) \mid a \in A, b \in A \text{ and } a \in b \,\}.$$

- The **identity relation** on A, denoted Id_A, is defined by

$$\mathrm{Id}_A \stackrel{\text{def}}{=} \{\, (a, b) \mid a \in A, b \in A \text{ and } a \doteq b \,\}.$$

Definition [2.]2.12

Let A and B be sets. The **Cartesian product** of A and B, denoted $A \times B$, is defined by

$$A \times B \stackrel{\text{def}}{=} \{\, (a, b) \mid a \in A \text{ and } b \in B \,\}.$$

Relations

Question

Let A and B be sets. Is $A \times B$ a set?

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Let A and B be sets. Is $A \times B$ a set?

Answer

Definition of $A \times B$ without using the textbook conventions.

$$A \times B \stackrel{\text{def}}{=} \{\, w \in \mathcal{P}(\mathcal{P}(A \cup B)) \mid w \doteq (a, b) \text{ for some } a \text{ and } b \text{ such that } \mathbf{P}(a, b) \,\},$$

where $\mathbf{P}(x, y)$: "$x \in A$ and $y \in B$".

That is, $A \times B$ is a set defined from the Axiom Scheme of Comprehension.

Relations

Notation

$A^2 \stackrel{\text{def}}{=} A \times A$.

Relations

Notation

$A^2 \stackrel{\text{def}}{=} A \times A$.

Remark

Note that a binary relation R is in A if and only if $R \subseteq A^2$.

Relations

Definition (p. 22)

Let A, B and C be sets. The **Cartesian product** of A, B and C, denoted $A \times B \times C$, is defined by

$$\begin{aligned} A \times B \times C &\stackrel{\text{def}}{=} (A \times B) \times C \\ &\doteq \{\, (a, b, c) \mid a \in A, b \in B \text{ and } c \in C \,\}. \end{aligned}$$

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Notation

$A^3 \stackrel{\text{def}}{=} A \times A \times A$.

Definition [2.]2.13

A set R is a **ternary relation** if all elements of R are ordered triples.

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Definition [2.]2.13

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A set R is a **ternary relation in** A if and only if $R \subseteq A^3$.

†See the corrections to the textbook.

Relations

Definition (p. 22)

Let A be a set. A **unary relation in A** is any subset of A.

Outline

Functions

Definition [2.]3.1

A binary relation F is a **function** (**mapping** or **correspondence**) if

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aFb_1 and aFb_2 imply $b_1 \doteq b_2$ for any a, b_1 and b_2.

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for each a in $\operatorname{dom} F$ there is exactly one b such that aFb.

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Example

Whiteboard.

Functions

Notation (p. 24)

$F : A \to B \overset{\text{def}}{=} F$ is a function with $\operatorname{dom} F \doteq A$ and $\operatorname{ran} F \subseteq B$.

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Notation (p. 24)

If F is a function with $\operatorname{dom} F$ we shall also use the following notations for the function F:

(a) $\langle F(a) \mid a \in A \rangle$,
(b) $\langle F_a \mid a \in A \rangle$ or
(c) $\langle F_a \rangle_{a \in A}$.

Functions

Discussion

Quote from the textbook:

> *"Function, as understood in mathematics, is a procedure, a rule, assigning to any object a from the domain of the function a unique object b, the value of the function at a." (p. 28)*

Is there any (implicit or explicit) procedure, rule or formula in our definition of function?

Functions

Remark

The current definition of a (one-value) function (of a real variable) is from Dirichlet who in 1837 wrote:

> *"If a variable y is so related to a variable x that whenever a numerical value is assigned to x, there is a rule according to which a unique value of y is determined, then y is said to be a* ***function*** *of the independent variable x."*
> *(Merzcbach and Boyer [1968] 2011, p. 452).*

Functions

Remark

The current definition of a function on arbitrary sets is from Cantor 1895:

> *"By a **covering of the aggregate** N **with elements of the aggregate**, M or, more simply, by a **covering of** N **with** M, we understand a law by which with every element n of N a definite element of M is bound up, where one and the same element of M can come repeatedly into application. The element of M bound up with n is, in a way, a one-valued function of n, and may be denoted by $f(n)$; it is called a **covering function of** n. The corresponding covering of N will be called $f(N)$." (Cantor [1895] 1915, p. 94)*

Lemma [2.]3.2 (function extensionality)

Let F and G be functions. $F \doteq G$ if and only if $\operatorname{dom} F \doteq \operatorname{dom} G$ and $F(x) \doteq G(x)$, for all $x \in \operatorname{dom} F$.

Functions

Definitions [2.]3.3

Let F be a function and A and B sets.

(a) F is a function **on** (**from**) A if and only if $\operatorname{dom} F \doteq A$.

(b) F is a function **into** (**to**) B if and only if $\operatorname{ran} F \subseteq B$.

(c) F is a function **onto** B if and only if $\operatorname{ran} F \doteq B$.

(d) The **restriction** of F to A, denoted $F \restriction A$, is the function defined by

$$F \restriction A \stackrel{\text{def}}{=} \{\, (x, y) \in F \mid x \in A \,\}.$$

Functions

Definition [2.]3.7

A function F is **one-to-one** (or **injective**) if for each $y \in \operatorname{ran} F$ there is only one x such that xFy. In other words, if $x_1, x_2 \in \operatorname{dom} F$ and $x_1 \dot{\neq} x_2$ implies $f(x_1) \dot{\neq} f(x_2)$.

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Example

Whiteboard.

Functions

Definition [2.]3.10

(a) Functions f and g are **compatible** if $f(x) \doteq g(x)$ for all $x \in \operatorname{dom} f \cap \operatorname{dom} g$.

(b) A set of functions F is a **compatible system of functions** if any two functions f and g from F are compatible.

Functions

Definition [2.]3.13

Let A and B be sets. The **set of functions** on A into B, denoted B^A, is defined by

$$B^A \stackrel{\text{def}}{=} \{ F \mid F : A \to B \}.$$

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Example

- $\{0,1\}^{\mathbf{N}}$: The set of infinity binary sequences.

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- $\emptyset^A \doteq \emptyset$ for $A \not\doteq \emptyset$ (no function can have a non-empty domain and an empty range).

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Example

- $\{0, 1\}^{\mathbf{N}}$: The set of infinity binary sequences.
- $\emptyset^A \doteq \emptyset$ for $A \not\doteq \emptyset$ (no function can have a non-empty domain and an empty range).
- $A^\emptyset \doteq \{\emptyset\}$ for any set A ($\emptyset$ is the only function with an empty domain).

Functions

Question

Let A and B be sets. Is B^A a set?

Functions

Question

Let A and B be sets. Is B^A a set?

Answer

Definition of B^A without using the textbook conventions.

$$B^A \stackrel{\text{def}}{=} \{ F \in \mathcal{P}(A \times B) \mid F : A \to B \}.$$

That is, B^A is a set defined from the Axiom Scheme of Comprehension.

Functions

Convention (p. 27)

Let $S = \langle S_i \mid i \in I \rangle$ be a function with domain I.

> *"We call the function $\langle S_i \mid i \in I \rangle$ an* ***indexed system of sets****, whenever we wish to stress that the values of S are sets."*

Functions

Definition (p. 27)

Let $S = \langle S_i \mid i \in I \rangle$ be an indexed system of sets. The **product** (or **generalised product**) of S, denoted $\prod S$, is defined by

$$\begin{aligned}\prod S &\stackrel{\text{def}}{=} \{\, f \mid f \text{ is a function on } I \text{ and } f_i \in S_i, \text{for all } i \in I \,\} \\ &\doteq \left\{\, f \;\middle|\; f : I \to \bigcup\{S_i \mid i \in I\} \text{ and } f_i \in S_i, \text{for all } i \in I \,\right\}.\end{aligned}$$

Functions

Example (informal)

Let A and B be two sets, $I = \{0, 1\}$ and let $S = \langle S_i \mid i \in I \rangle$ be an indexed system of sets where $S_0 = A$ and $S_1 = B$. Then

$$\prod S \doteq \left\{ f \;\middle|\; f : I \to \bigcup\{A, B\}, \text{ such that } f_0 \in A \text{ and } f_1 \in B \right\}.$$

Now, we can define the one-to-one correspondence

$$h : \prod S \to A \times B$$
$$h(f) = (f_0, f_1).$$

Functions

Remark

Let $\langle S_i \mid i \in I \rangle$ be an indexed system of sets. If $S_i = B$ for all $i \in I$, then

$$\begin{aligned} \prod S &\doteq B^I \\ &\doteq \{\, f \mid f : I \to B \,\}. \end{aligned}$$

Functions

Notation (p. 27)

Let $\prod S$ be the product of an indexed system of sets $S = \langle S_i \mid i \in I \rangle$. We shall also use the following notations for the product $\prod S$:

(a) $\prod \langle S_i \mid i \in I \rangle$,

(b) $\prod\limits_{i \in I} S(i)$ or

(c) $\prod\limits_{i \in I} S_i$.

Functions

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Functions

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Let $\prod S$ be the product of an indexed system of sets $S = \langle S_i \mid i \in I \rangle$.
Is $\prod S$ a set?

Answer

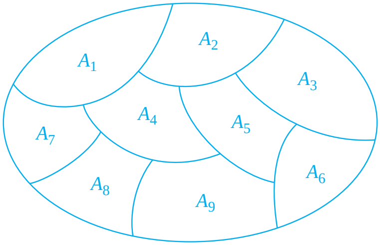
Definition of $\prod S$ without using the textbook conventions.

$$\prod S \stackrel{\text{def}}{=} \left\{ f \in \mathcal{P}(I \times \bigcup\{S_i \mid i \in I\}) \;\middle|\; f \text{ is a function on } I \text{ and } f_i \in S_i, \text{for all } i \in I \right\}$$

That is, $\prod S$ is a set defined from the Axiom Scheme of Comprehension.

Outline

Equivalence and Partitions

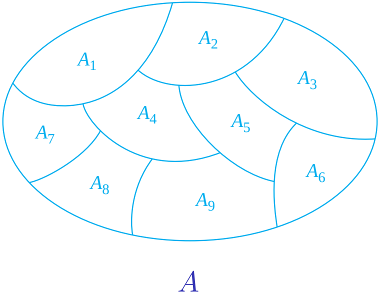


A

(a) $A_i \cap A_j = \emptyset$ for $i \neq j$,

(b) $A = \bigcup A_i$.

Equivalence and Partitions[†]



(a) $A_i \cap A_j = \emptyset$ for $i \neq j$,

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[†] Figure source: (Rosen [1988] 2012, Fig. § 9.5, Fig. 1).

Equivalence and Partitions

Definition [2.]4.1

Let R be a binary relation in a set A. The relation R is

(a) **reflexive in** A if for all $x \in A$, xRx;

(b) **symmetric in** A if for all $x, y \in A$, xRy implies yRx.

(c) **transitive in** A if for all $x, y, z \in A$, xRy and yRz imply xRz;

(d) an **equivalence on** A if R is reflexive, symmetric and transitive in A.

Equivalence and Partitions

Questions

(a) Let $A = \{a, e, i, o, u\}$. Is the identity relation in A an equivalence on A?

Equivalence and Partitions

Questions

(a) Let $A = \{a, e, i, o, u\}$. Is the identity relation in A an equivalence on A?

(b) Let $A \not\doteq \emptyset$ be a set. Is the relation $\emptyset$ an equivalence on A? What about if $A \doteq \emptyset$?

Equivalence and Partitions

Questions

(a) Let $A = \{a, e, i, o, u\}$. Is the identity relation in A an equivalence on A?

(b) Let $A \not\doteq \emptyset$ be a set. Is the relation $\emptyset$ an equivalence on A? What about if $A \doteq \emptyset$?

(c) Let $A \not\doteq \emptyset$ be a set. Is the relation $A \times A$ an equivalence on A? What about if $A \doteq \emptyset$?

Equivalence and Partitions

Questions

(a) Let $A = \{a, e, i, o, u\}$. Is the identity relation in A an equivalence on A?

(b) Let $A \not\doteq \emptyset$ be a set. Is the relation $\emptyset$ an equivalence on A? What about if $A \doteq \emptyset$?

(c) Let $A \not\doteq \emptyset$ be a set. Is the relation $A \times A$ an equivalence on A? What about if $A \doteq \emptyset$?

(d) Let A be a singleton. It is possible to define an equivalence on A?

Definition [2.]4.3

Let E be an equivalence on A and let $a \in A$. The **equivalence class of** a **modulo** E, denoted $[a]_E$, is the set defined by

$$[a]_E \stackrel{\text{def}}{=} \{\, x \in A \mid xEa \,\}.$$

Definition [2.]4.5

A system S of non-empty sets is called a partition of A if

(a) S is a system of mutually disjoint sets, i.e., if $C, D \in S$ and $C \mathrel{\dot{\neq}} D$, then $C \cap D \doteq \emptyset$,

(b) the union of S is the whole set A, i.e., $\bigcup S \doteq A$.

Equivalence and Partitions

Definition [2.]4.8

Let S be a partition of A. The relation E_S in A is defined by

$$E_S = \{\, (a,b) \in A \times A \mid \text{ there is } C \in S \text{ such that } a \in C \text{ and } b \in C \,\}.$$

Equivalence and Partitions

Definition [2.]4.8

Let S be a partition of A. The relation E_S in A is defined by

$$E_S = \{\, (a,b) \in A \times A \mid \text{ there is } C \in S \text{ such that } a \in C \text{ and } b \in C \,\}.$$

Remark

Note that objects a and b are related by E_S if and only if they belong to the same set from the partition S.

Equivalence and Partitions

Theorem [2.]4.9

Let S be a partition of A; then the relation E_S is an equivalence on A.

Outline

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